在定量社会科学研究中负责任地使用人工智能

社会研究方法课程单元

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研究方向：运用因果推断与计算方法评估数字信息技术的社会与政治影响

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本次课的目标

（……（计算技能）门槛降低……专业知识的重新分配……）

“对学生而言，问题不再是他们能否编程，而是他们能否审视、批判、调整并负责任地部署 AI 辅助方法，而不把便利误当作……”(Zhou et al., 2026)

the ease of generation raises the value of human judgment in CCR further, necessitating sharper and more transparent human oversight.

This redistribution of expertise has major implications for pedagogy and training. For instructors, GenAI unsettles the sequencing of computational education itself: it is no longer obvious whether CCR methods courses should begin with traditional techniques and only later introduce generative systems, or whether GenAI should be integrated from the outset as part of the field's ordinary methodological environment. For students, the question is no longer whether they can code, but whether they can audit, critique, adapt, and responsibly deploy AI-assisted methods without mistaking convenience

for competence.

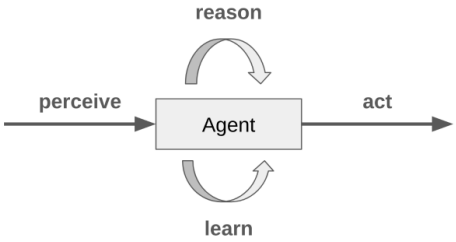
At the same time, the seeming democratization of computational workflows may conceal new inequalities. Well-resourced scholars with strong methodological priors are often best positioned to benefit from GenAI because they can afford to access the most advanced models and the computational power to run them. Furthermore, they may be better equipped to

来源：Zhou et al. (2026)

引言

1.1 什么是人工智能?

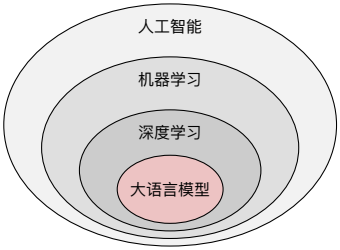
- 能够智能地行动的计算智能体 (Poole & Mackworth, 2023)
- 但什么是智能? ——见右图



来源: Stanford CS221, Percy Liang

1.1 什么是人工智能?

- 能够智能地行动的计算智能体 (Poole & Mackworth, 2023)
- 但什么是智能? ——见右图
- 更重要的是，请注意以下概念之间的区别 ……



1.2 大语言模型简史

- * 人工智能是一个已发展数十年的古老领域 (参见 Poole & Mackworth, 2023)
- * 前大模型时代，早期工具主要基于人类产生的痕迹数据进行分类、聚类或预测 (Zhou et al., 2026)。[a]
- * 大语言模型的诞生：Transformer (Vaswani et al., 2017)，一种可并行训练的模型结构，使大规模数据与大模型成为可能
- 后大模型时代：本课聚焦 2020 年后的大语言模型时代 (以 GPT-3 为标志； Xu et al., 2026) 及其对计算社会科学研究的意义；并简要演示基于 RoBERTa 的文本分类

[a]类似地，Percy Liang 以 2016 年、2018 年和当下划分三个时期，见 https://cs336.stanford.edu/lectures/?trace=lecture_01

Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples. Recent work has achieved significant improvements in computational efficiency through factorization tricks [18] and conditional computation [26], while also improving model performance in case of the latter. The fundamental constraint of sequential computation, however, remains.

Attention mechanisms have become an integral part of compelling sequence modeling and transduction models in various tasks, allowing modeling of dependencies without regard to their distance in the input or output sequences [2, 16]. In all but a few cases [22], however, such attention mechanisms are used in conjunction with a recurrent network.

In this work we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs.

2 Background

The goal of reducing sequential computation also forms the foundation of the Extended Neural GPU [20], ByteNet [15] and ConvS2S [8], all of which use convolutional neural networks as basic building block, computing hidden representations in parallel for all input and output positions. In these models, the number of operations required to relate signals from two arbitrary input or output positions grows in the distance between positions, linearly for ConvS2S and logarithmically for ByteNet. This makes it more difficult to learn dependencies between distant positions [11]. In the Transformer this is reduced to a constant number of operations, albeit at the cost of reduced effective resolution due to averaging attention-weighted positions, an effect we counteract with Multi-Head Attention as described in section 3.2.

Self-attention, sometimes called intra-attention is an attention mechanism relating different positions of a single sequence in order to compute a representation of the sequence. Self-attention has been used successfully in a variety of tasks including reading comprehension, abstractive summarization, textual entailment and learning task-independent sentence representations [4, 22, 23, 19].

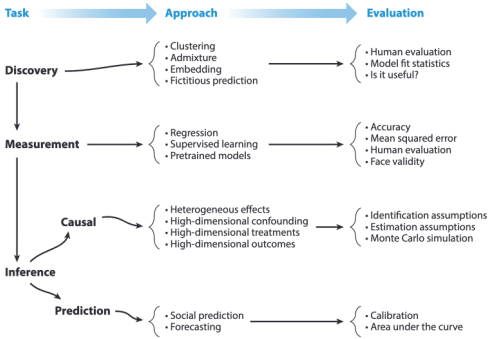
End-to-end memory networks are based on a recurrent attention mechanism instead of sequence-aligned recurrence and have been shown to perform well on simple-language question answering and

摘自 Vaswani et al. (2017)

前大模型时代的 AI

2.1 前大模型时代 AI 在社会科学中的应用

- 按照 Grimmer et al. (2021) 的分类，社会科学研究常借助机器学习完成多种任务：
  - 发现；例如文本分类 （教程一）
  - 测量；例如回归
  - 因果推断；例如异质性处理效应
  - 预测；例如预测社会趋势
- 如有兴趣，可选修汇丰商学院 FIN528 金融机器学习 / 高级机器学习



来源：Grimmer et al. (2021)（第 23 页）

*TUT1

使用 RoBERTa 进行情感分类

在线材料：

Google Colab 链接

大语言模型

3.1 大语言模型在计算社会科学中的角色

- 大语言模型（LLM）大致指参数量不少于 10 亿的语言模型（Percy Liang；如 GPT、Claude）[1]
- 一篇近期综述提出大语言模型在计算社会科学中的四种角色：研究主体、助手、合作者、评估者

Table 1 | Representative recent work leveraging LLMs in social scientific study, organized by the research roles they have

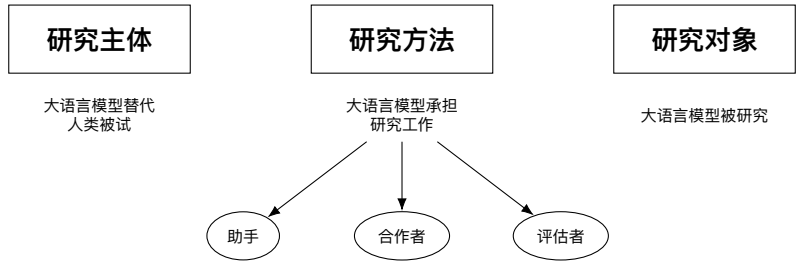
Research role	Description	Example works
Research subjects	Generating survey responses	Refs. 32,43,48
	Simulating behaviour sequences	Refs. 52,53,55
	Simulating social interaction	Ref. 61,63,65,67
Research assistants	Interviewing human participants	Refs. 44,73,74,76
	Supporting experiments	Refs. 81–83
	Analysing textual content	Refs. 85,93,99,102
	Mining social networks	Refs. 107–109
Research collaborators	Reviewing literature	Refs. 117,120,122
	Proposing hypotheses	Refs. 126–128
	Designing research protocols	Refs. 129,130
Research evaluators	Evaluating research protocols	Refs. 132,133,137,140
	Critiquing research outcomes	Refs. 141,142,144

来源：Xu et al. (2026)

[1]本课并非计算机课程，采用这一定义应无大碍。

3.1 大语言模型在计算社会科学中的角色

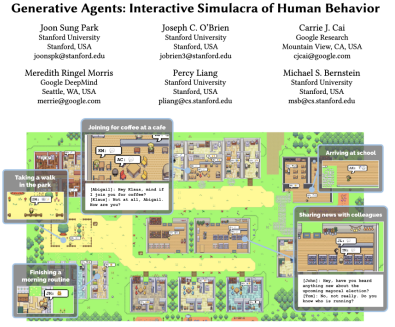
- 从更多视角出发，我们可以对大语言模型在计算社会科学中的角色提出更全面的分类；后续幻灯片将逐一展开



来源：四种角色（研究主体、助手、合作者、评估者）的分类见 Xu et al. (2026)；四个类别见 Hai Liang (2026)；两种角色（研究工具、研究对象）改编自 Zhou et al. (2026)

3.2 大语言模型作为研究主体：类别

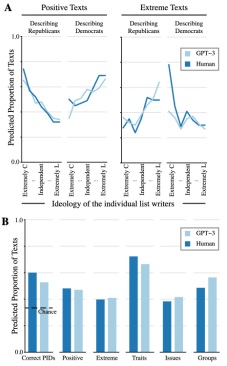
- 根据 Xu et al. (2026)，大语言模型可充当研究主体：
  - 生成问卷回答 (Dillion et al., 2023; Dominguez-Olmedo et al., 2024; Hämäläinen et al., 2023; Y. Xie & Xie, 2025)
  - 模拟行为序列 (Park, O'Brien, et al., 2023; L. Wang et al., 2025; Yan et al., 2024)
  - 模拟社会互动 (Ashery et al., 2025; Gao et al., 2025; C. Xie et al., 2024; Zhang et al., 2024)



模拟小镇中的生成式智能体；Park, O'Brien, et al. (2023)

3.3 大语言模型作为研究主体：优势与不足

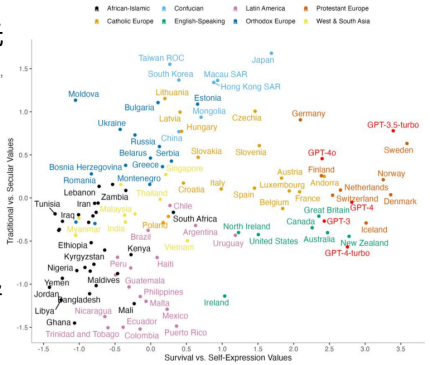
- **优势：**
  - 在问卷调查中成功复现人类回答 (Argyle, Busby, et al., 2023)
  - 产生可信的个体行为与涌现的社会行为 (Park, O'Brien, et al., 2023)
  - 复现社会规范的涌现 (Ashery et al., 2025)
  - 等等



GPT-3 与 Lucid 调查中人类回答的对比分析；Argyle, Busby, et al. (2023) 图 3

3.3 大语言模型作为研究主体：优势与不足

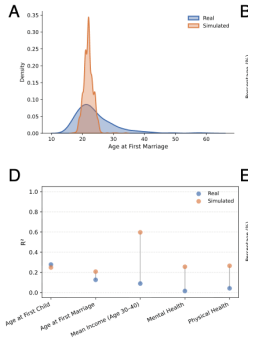
- **不足：**
  - 难以反映“难以触达的人群”(Bollen et al., 2026)；无法捕捉经验与文化层面的深度 (Pataranutaporn et al., 2025)
  - 体现特定的党派 (Santurkar et al., 2023) 或文化偏差 (Tao et al., 2024)
  - 接近均值但变异不足 (Bisbee et al., 2024)；在意见或态度研究中存在强烈且多变的偏差、方差偏低 (Boelaert et al., 2025)；容易同时误描并扁平化人口群体特征 (A. Wang et al., 2025)；并倾向于将异质性压缩为简化的类型结构 (Y. Xie & Xie, 2025)



大语言模型生成的价值观回答与不同国家相比存在偏离；Tao et al. (2024) 图 1

3.3 大语言模型作为研究主体：优势与不足

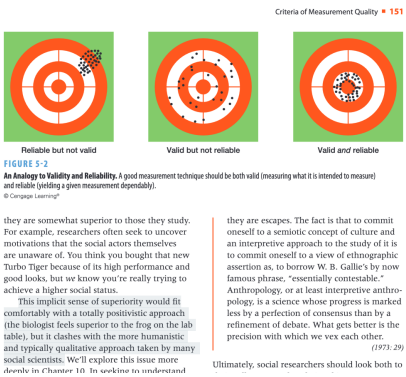
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未能捕捉真实世界分布的离散程度；Y. Xie & Xie (2025) 图 3

3.4 大语言模型作为研究主体：评估

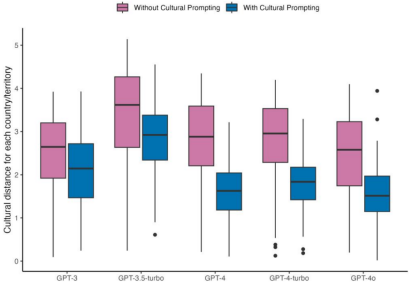
- 效度与信度仍是关注焦点 (Zhou et al., 2026)；请参阅叶老师此前的课程



效度与信度；见 Babbie (2021)

3.5 大语言模型作为研究主体：改进

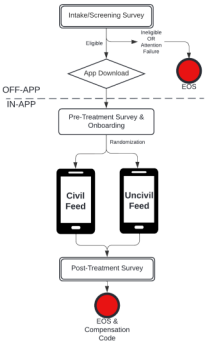
- 效度与信度仍是关注焦点 (Zhou et al., 2026)
- 三种改进策略
  - 对齐（使用前）：调整模型输出；主要由企业和计算机科学研究完成
  - 优化（使用中）：通过**微调或提示**策略提升特定任务的表现 (如 Kolluri et al., 2025; Tao et al., 2024; Y. Xie & Xie, 2025)
  - 校正（使用后）：事后修正；计量经济学与机器学习文献中的新兴方向 (如 Chernozhukov et al., 2018; Egami et al., 2024)



提示可缓解文化偏差；来源：Tao et al. (2024) 图 2

3.6 大语言模型作为研究方法：助手

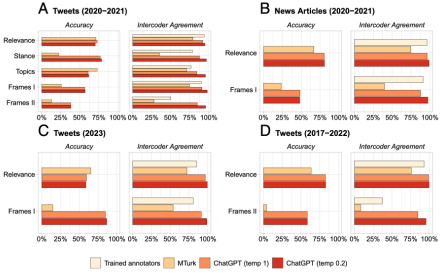
- 访谈人类参与者（质性研究）(Hanschmann et al., 2025; H. Li et al., 2024; Park et al., 2026; Tuggener et al., 2024)
- **支持实验研究** (Allamong et al., 2026; Argyle, Bail, et al., 2023; Hackenburg & Margetts, 2024; Velez & Liu, 2025)
- 分析文本内容 (Chew et al., 2023; Gilardi et al., 2023; Goyal et al., 2023; Jiang et al., 2024)
- 挖掘社交网络 (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



随机实验中的合成传播环境；Allamong et al. (2026)

3.6 大语言模型作为研究方法：助手

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- 挖掘社交网络 (Feng et al., 2024; Luceri et al., 2024; Yu et al., 2025)



标注质量优于众包工作者；Gilardi et al. (2023)

TUT2

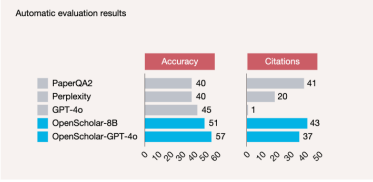
使用大语言模型进行主题分类

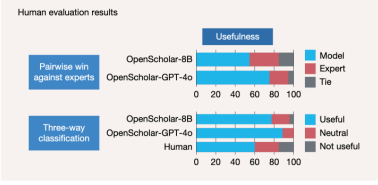
别忘了效度检验！

在线材料：GitHub

3.7 大语言模型作为研究方法：合作者

- **文献综述** (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
- 提出假设 (Si et al., 2024; Q. Wang et al., 2024; Yang et al., 2024)
- 设计研究方案 (Baek et al., 2025; Manning et al., 2024)

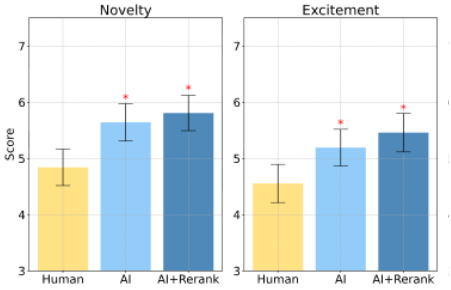




由大语言模型完成的博士水平文献综述系统；见 Asai et al. (2026)

3.7 大语言模型作为研究方法：合作者

- 文献综述 (Asai et al., 2026; Dennstädt et al., 2024; Gehrmann et al., 2024; Jiang et al., 2024)
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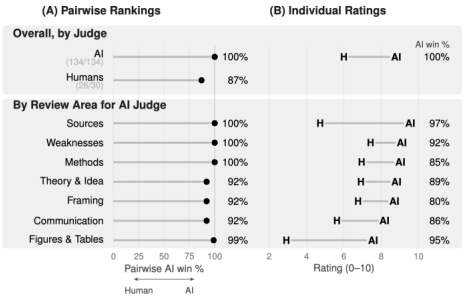
大语言模型生成的假设在新颖性与吸引力上更优；见 Si et al. (2024)

TUT3

数据分析中的智能体建模
在线材料：GitHub

3.8 大语言模型作为研究方法：评估者

- 评估研究方案 (Lin & Chen, 2023; Shankar et al., 2024; X. Wang et al., 2024; Zheng et al., 2023)
- **评议研究成果** (Ke et al., 2024; King & Worthington, 2026; M. Li et al., 2024; Liang et al., 2024)



AI 评审全面优于人类评审；来源：(King & Worthington, 2026)；未经同行评议，但设计严谨

TUT4

批判性评估大语言模型生成的审稿意见

GitHub

3.9 大语言模型作为研究对象

- “一种重塑媒介系统中生产、流通、阐释与权力的传播技术，引发了关于真实性、可及性以及社会不平等再生产的关键问题。”(Zhou et al., 2026)
- 例如，Liu and Zhang (2026) 评估了大语言模型生成的摘要如何塑造公众情绪，其处理变量即为模型本身。

RQ2. Whether and how do the emotions of AIGS influence sentiment in subsequent posts associated with the hashtag

To explore the effect of emotions in AIGS on the subsequent public sentiment, we used a Difference-in-Difference (DiD) design in natural experiments on Weibo. Before the DiD analysis, we conducted an event study plot on sentiment scores to examine the parallel assumption. As Figure 3 shows[2], the parallel trends assumption was met, indicating that there is no evidence of systematic platform-level selection bias between hashtags with and without the AIGS feature.

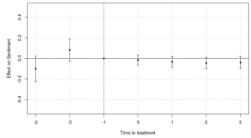


Figure 3: Event Study Plot on Sentiment Scores (Treatment versus Control)

Liu & Zhang (2026) 中作为研究对象的 AI 生成摘要（AIGS）

小测验

在线上群组中提出一个以大语言模型为研究对象的研究问题
请简要说明：

（a）关注的问题

（b）变量的操作化

（c）最重要的一张图

（d）预期贡献

3.10 * 大语言模型用于推断：新前沿

- 双重/去偏机器学习在用灵活的机器学习模型估计冗余函数的同时，仍能实现有效的因果推断 (Chernozhukov et al., 2017)；在此基础上，Imai and Nakamura (2026) 利用生成式 AI 的内部表征，估计文本、图像等非结构化处理变量的因果效应



PNAS RESEARCH ARTICLE | SOCIAL SCIENCES

Leveraging generative AI for causal inference with unstructured data

Kosuke Imai[a,1,2] and Kentaro Nakamura[a,1]

Affiliations are included on p. 5.

Edited by David M. Blei, Columbia University, New York, NY; received October 24, 2025; accepted July 29, 2026 by Editorial Board Member Kenneth W. Wachter

We introduce GenAI-Powered Inference (GPI), a statistical framework for causal inference using unstructured data, including text and images. GPI leverages open-source pretrained Generative AI (GenAI) models—such as large language models and diffusion models—not only to generate unstructured data at scale but also to extract low-dimensional representations that are guaranteed to capture their underlying structure. Applying machine learning to these representations, GPI enables estimation of causal effects while quantifying estimation uncertainty. Unlike existing approaches to representation learning, GPI does not require fine-tuning of GenAI models, making it computationally efficient and broadly accessible. We illustrate the versatility of the GPI framework through three applications: 1) estimating the effects of Chinese social media censorship while adjusting for textual confounders, 2) isolating the impact of specific image features from that of other correlated features in the same image, and 3) assessing the persuasiveness of political rhetoric. An open-source software package is available for implementing GPI.

generative AI | causal inference | machine learning | unstructured data

The emergence of Generative AI (GenAI), including large language models and diffusion models, has transformed society and scientific research. In this paper, we show that GenAI can also serve as a powerful tool for causal inference with unstructured data such as text and images. Specifically, we introduce a statistical framework, GenAI-Powered Inference (GPI), which leverages open-source GenAI models to improve causal inference with high-dimensional unstructured data.

We illustrate the power of GPI through the following three applications:

1. **Text as confounder**: Estimating the causal effects of a treatment variable by adjusting for latent confounders embedded in textual data
2. **Image as Treatment**: Isolating the impact of specific image features from that of other correlated features in the same image
3. **Structural model of text**: Fitting a structural model that incorporates both observed and latent textual features as covariates

A common challenge in these settings is that while high-dimensional unstructured data are observed, we do not know, a priori, which features of such data serve as relevant confounders.

GPI addresses this challenge by leveraging the ability of GenAI to generate or reproduce unstructured data and extracting lower-dimensional vector representations that are guaranteed to contain all the relevant information. Machine learning is then applied to these representations to estimate a *deconfounder*, which is a summary of confounding

Significance

We introduce GenAI-Powered Inference (GPI), a unified methodological framework that integrates open-source large language and diffusion models into causal inference. GPI uses generative AI (GenAI) models to extract low-dimensional representations that retain the information necessary to generate complex unstructured data. This enables researchers to estimate causal effects without directly modeling high-dimensional objects or fine-tuning foundation models. We show that estimating and adjusting for a deconfounder leads to valid causal inference with unstructured data. By leveraging pretrained GenAI models while avoiding restrictive modeling assumptions, GPI delivers strong empirical performance across diverse settings. We demonstrate the versatility of the framework in three distinct applications and provide open-source software to facilitate its use in applied research.

来源：Imai & Nakamura (2026)

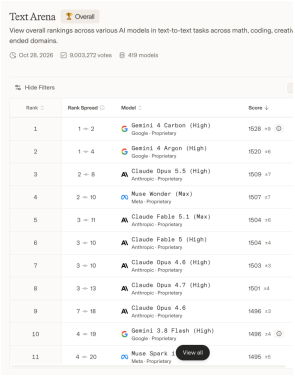
负责任地使用

4.1 负责任地使用大语言模型：模型选择

- 将模型视为一个函数 $f(data, algo, object)$，各部分的差异会显著塑造其领域内能力（如更擅长编程或写作）
- 因此，我们可能需要根据不同模型的优势与局限，为不同任务选择不同模型

4.1 负责任地使用大语言模型：模型选择

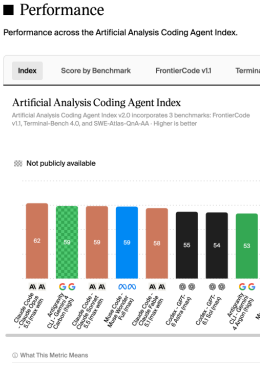
- 例如，Gemini Carbon 可能更擅长写作任务，而 Claude Opus 5.5 在工程任务上表现更好



来源：https://arena.ai/leaderboard/chat/text（2026 年 11 月 3 日）

4.1 负责任地使用大语言模型：模型选择

- 例如，Gemini Carbon 可能更擅长写作任务，而 Claude Opus 5.5 在工程任务上表现更好
- 总体而言，各模型厂商的商业化路径不同，面向工程任务优化商业模式的厂商可能优先发展编程能力（如 Anthropic）；使用某一模型前，请查看基准测试、回顾文献或进行预测试



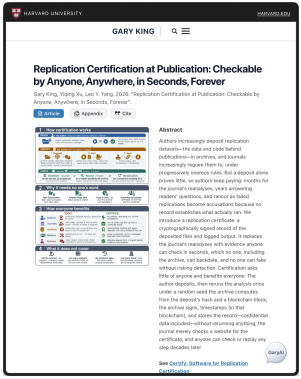
来源：
https://artificialanalysis.ai/agents/coding-agents#coding-agents-index
(2026 年 11 月 3 日)

4.2 负责任地使用大语言模型：效度与信度

- 再次强调：在研究中使用模型输出之前，务必检验其效度与信度

4.3 负责任地使用大语言模型：可复现性

- 可复现性对科学至关重要
- 通过以下方式确保研究中人工智能部分可复现：
  - 优先使用 API 调用，而非聊天界面
  - 记录模型版本、提示词、思考等级与参数
  - 提供详细的操作流程
  - 共享代码与数据
  - 固定随机种子与温度（具体请自行检索）



发表时的复现认证；来源：King et al. (2026)

*TUT5 & 6

多模态目标检测与受控刺激材料生成
在线材料：GitHub

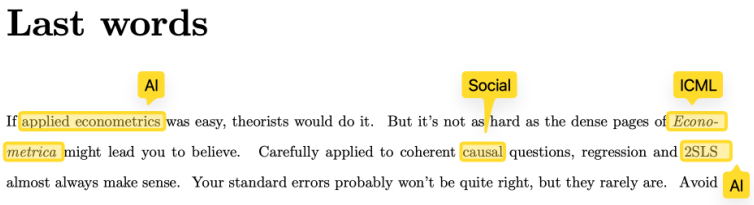
要点总结

5.1 要点总结

- AI 模型（包括大语言模型）只是高维数据分析的工具；在定量社会研究中，它们可以充当研究主体、研究对象和研究方法
- 作为研究方法，应以负责任的方式使用，审慎考虑效度、信度与可复现性
- 与其他方法一样，尽管它们发展迅速，其使用仍应建立在我们对研究、业界与生活中好问题的品味之上，而这种品味始终是高等教育的核心

5.1 要点总结

- “……保持怀疑……但别慌！”(Angrist & Pischke, 2009)



来源：Angrist & Pischke (2009)

资源

6.1 资源

- **为任务找到最合适的 AI 模型**
  - Artificial Analysis https://artificialanalysis.ai/
  - Arena.ai，AI 模型盲测排行榜 https://arena.ai/leaderboard/
- **发现稿件中的问题**
  - Refine，发表级技术审阅；已被美国经济学会（AEA）采用 https://www.refine.ink/
  - OpenAIReview，芝加哥大学 HAI 开源的多智能体稿件审阅方案 https://github.com/ChicagoHAI/OpenAIReview
- **复核参考文献与引用**
  - CiteVerify，“一款用于核验论文参考文献的本地工具”，https://github.com/cuihuash/CiteVerify
  - 与前沿模型对话也会有所帮助

6.1 资源

- **统一管理多个 AI 模型与服务**
  - macOS：magpie，Mac 本地 AI 路由管理工具 https://github.com/yetone/magpie
  - 其他平台：cc-switch，https://github.com/farion1231/cc-switch
- **用母语阅读科学文献**
  - PDFMathTranslate，以你的语言阅读科学 PDF https://github.com/PDFMathTranslate/PDFMathReader (Ouyang et al., 2025)

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